

FAST-NUCES LAHORE**Fall 2025***Section: BCS 1B**Due Date: 9th Oct, 2025***ROLL NO.** _____ **NAME:** _____**Instructions:**

1. Assignment should be hand-written. Typed assignments will not be accepted.
2. Do not forget to mention Question Number.
3. Use A4-size pages.
4. Attach this page with your assignment pages.
5. Late submissions are not acceptable.

Question 01

A guitar string is plucked so that its midpoint oscillates with an amplitude of $1.00\ \mu\text{m}$.

- (a) At what frequency must the string vibrate so that the midpoint's maximum acceleration is equal to the acceleration due to gravity, $g = 9.8\ \text{m/s}^2$?
- (b) If the string instead vibrates at $10\ \text{kHz}$, what is the ratio of the midpoint's maximum acceleration to g ?
- (c) For very high frequencies, will the acceleration be much larger than g or much smaller? Briefly explain.

(15 marks)

Question 02

The displacement of a particle in simple harmonic motion is given by

$$x(t) = (8.0\ \text{cm}) \cos\left[(4\pi\ \text{rad/s})t + \frac{\pi}{6}\ \text{rad}\right].$$

At time $t = 1.5\ \text{s}$, determine:

1. The **displacement** of the particle,
2. The **velocity** of the particle,

3. The **acceleration** of the particle,
4. The **phase angle** of the motion at this instant,
5. The **frequency** and **period** of the motion.

(20 marks)

Question 03

A transverse sinusoidal wave travels along a taut string in the $+x$ direction with phase speed $v = 72$ m/s. At $t = 0$, the element of string at $x = 0.150$ m has a transverse displacement $y = -2.5$ cm and is not moving. The maximum transverse speed of a string element at $x = 0$ is 16 m/s.

If the wave is written in the form

$$y(x, t) = y_m \sin(kx \pm \omega t + \phi),$$

determine:

- (a) the frequency f ,
- (b) the wavelength λ ,
- (c) the amplitude y_m ,
- (d) the wave number k ,
- (e) the angular frequency ω ,
- (f) the phase constant ϕ , and
- (g) the correct choice of sign in front of ω (justify briefly).

(25 marks)

Question 04

A block of mass 0.300 kg is placed on a *frictionless* horizontal surface and connected to two identical springs, one on each side, each having a spring constant of 6200 N/m.

1. Derive the **effective spring constant** of the system.
2. Determine the **angular frequency** of oscillation.
3. Find the **frequency** in hertz and the **period** of oscillation.
4. If the amplitude of oscillation is 2.0 cm, what is the **maximum acceleration** of the block?

(20 marks)

Question 05

An oscillator consists of a block of mass 0.400 kg attached to a horizontal spring. The block oscillates on a *frictionless* surface with an amplitude of 25.0 cm. It is observed that the oscillator completes five full oscillations in 2.00 s.

Find the following:

1. The **period** of the motion.
2. The **frequency** in hertz.
3. The **angular frequency** in radians per second.
4. The **spring constant** of the spring.
5. The **maximum speed** of the block.
6. The **maximum force** exerted by the spring on the block.

(12 marks)

Question 06

A transverse wave travels along a very long string. Its displacement (in *millimeters*) is

$$y(x, t) = (5.0 \text{ mm}) \sin\left(12\pi x - 36\pi t + \frac{\pi}{4}\right),$$

where x is in meters and t is in seconds.

Determine:

1. the **amplitude** (express your answer in centimeters);
2. the **wavelength** (in centimeters);
3. the **frequency**;
4. the **wave speed**;
5. the **direction of propagation**;
6. the **maximum transverse speed** and the **maximum transverse acceleration** of a string element;
7. the **transverse displacement** y and the **transverse velocity** $\partial y / \partial t$ at $x = 0.250 \text{ m}$, $t = 0.125 \text{ s}$;
8. the **smallest positive time** at which the element at $x = 0$ first reaches its **maximum positive displacement**.

(13 marks)